

User Perspectives on the Role of LLM-based AI Agents in Social VR

Alexandra Kitson*
University of Victoria

Mirjana Prpa†
Northeastern University

ABSTRACT

As Large Language Model (LLM)-based AI agents emerge in social Virtual Reality (VR), understanding the roles they might play is critical. We conducted interviews in VRChat with experienced users and developers (N=6) and, using thematic analysis, identified five primary roles: information sharing, personal companionship, moderation, content generation, and self-development and learning. While participants recognized the potential for increased accessibility and support, they expressed concerns regarding over-moderation, the replacement of human creativity, and the psychological impact of AI co-dependency. These preliminary findings highlight the need to balance AI utility with the preservation of authentic human connection in social VR.

Index Terms: social virtual reality, embodied conversational agents, large language models

1 INTRODUCTION

Social Virtual Reality (VR) platforms like VRChat¹ have long served as digital hubs for human connection by elevating users' sense of presence and embodiment [3]. However, the integration of Large Language Model (LLM)-based AI agents is beginning to shift the landscape from purely human-to-human interaction to a hybrid social ecosystem [4, 8]. Unlike the static Non-Player Characters (NPCs), LLM-based agents can engage in fluid, context-aware, and emotionally resonant dialogue, offering a novel form of social presence within virtual environments [9].

As developers begin to integrate these agents into social VR, a critical tension arises between the potential for AI to enhance the user experience and the risk of it disrupting the human-centric nature of these spaces [1]. While existing research has explored the opportunities and challenges for AI agents in social VR through the analysis of scientific publications [5, 9, 7], there is a gap in understanding the perceptions of those at the forefront of this integration, i.e., the developers who build these agents and the power-users who interact with them daily. This paper addresses this gap by exploring the following **research question**: What roles could LLM-based AI agents play in social VR according to its users and developers?

2 METHODS

2.1 Study Design and Procedure

We conducted semi-structured interviews with N=6 participants: 2 non-binary, 4 men; age ($M = 34$, $SD = 10.35$ years); 3 Europe, 3 USA country of residence. We recruited participants through Reddit (r/VRchat) and Discord (Studio JAW; Joken's AI Friends; Open-Source VRChat AI). We asked for participants who create and/or interact with AI bots in VRChat and screened them via an online Qualtrics survey. Participants had either 1-5 years ($n=1$) or 5 years



Figure 1: VRChat player interacting with the AI Agent 'Joken'.

($n=5$) experience with VRChat, and all reported engaging with social VR every day. Three participants were developers of AI agents in VRChat (P3, P5, P6), and three had interacted with AI agents (P1, P2, P4), including AI Joken, Celeste-AI, Nemu AI, MapleAI, ArcticLucky AI, VRC_AI.Bot(Tigerbee), ClunkerAI, and Irrbot. All participants received a \$20 Amazon gift card for their time. Our institutional ethics review board approved this project.

We interviewed participants within social VR [6] in the VRChat world "Kit Café DAY" (see Figure 1). Each interview lasted an hour except for P6, which lasted two hours at their request. The interviewer, one of the authors, used a Meta Quest 3 headset that was tethered to an ASUS ROG Zephyrus G15 laptop running VRChat through SteamVR. We screen-recorded the interviews in VRChat using OBS Studio to ensure high-quality local data storage. We then transcribed the audio using Microsoft 365's transcription tool. Interview transcripts were imported into NVivo 14 for analysis.

2.2 Interview Questions

Questions were divided into three parts: introductions and context, guiding and ad hoc questions, and speculative questions about the future of AI agents in social VR. Interview questions: (1) What first got you interested in VRChat? (2) Tell me about your experiences with AI agents in VR. (3) In what kinds of situations do you think AI agents in social VR could be helpful or unhelpful? Why? (4) In the future, how would you like to see AI agents be part of social VR? Why? (5) Is there anything else you'd like to share with us?

2.3 Analysis

We conducted a thematic analysis on interview transcripts [2]. Two researchers, with extensive qualitative training, first watched all the interview videos and read through the transcripts. Next, we independently coded the data inductively in NVivo 14. We compared preliminary codes and discussed and agreed upon a set of codes through multiple rounds of coding. To ensure the robustness of our analysis, we cross-checked the transcription against the original video files. Next, we exported our codes to a shared Miro board where we grouped data into themes. We discussed and clearly defined these themes, ensuring each represented both the codes extracted and the entire dataset. Finally, we integrated our analysis into the following narrative, presenting the results and discussion.

*e-mail: akits@uvic.ca

†e-mail: m.prpa@northeastern.edu; new affiliation: Hong Kong University of Science and Technology, e-mail: mirjanaprpa@hkust-gz.edu.cn

¹<https://hello.vrchat.com>

3 RESULTS AND DISCUSSION

3.1 Information Sharing

Participants viewed AI agents as a tool for information sharing in social VR by taking the role of guide or personal assistant. Specifically, participants thought AI agents could be the first point of contact in VRChat to greet or guide players on social VR etiquette. For example, P02 said that *‘if I want to talk to you, I have to make sure player size and our eyes is to be on the same level...if you don’t do it, people will just ignore you’*. Participants also thought AI agents could be a customer care agent, assistant, and buddy finder. P05 expanded on the idea of a personal assistant: *‘you might want one that’s also not open, not visible to everyone, might be one, you can mute yourself and start talking to the bot, and then unmute yourself and the bot could be gone’*. Potentially, AI agents could help onboard newcomers: *‘improve accessibility for people that are not super familiar with that new way of interacting’* (P04).

3.2 Personal Companion

Participants stated that AI agents can act more like an intimate partner. In this role, an AI agent could be a clone of a friend, a companion, or even a sexual partner. Participants said that many VRChat players want someone to listen non-judgmentally, and that an AI might be able to do that without burdening an actual person. On the other hand, some participants were skeptical of the use of AI over real human connection: *‘do you really need a language model with artificial intelligence to make people feel less lonely? I don’t know. They need to believe it’s intelligent’* (P02). P04 was weirded out by the idea of *‘animating dead relatives’* using AI agents in social VR. From our conversations with VRChat users and developers, we had the sense of morally gray areas where people develop AI co-dependencies in a way that replaces real human interaction.

3.3 Moderator

Participants said AI agents in social VR could moderate, upholding VRChat’s terms of service while ensuring virtual events run smoothly, e.g., mitigating bullying and harassment. However, some participants were skeptical of an AI’s ability to moderate because of the unique social dynamics within communities that would be difficult to parse. Moreover, participants felt there is a danger in over-moderation, potentially kicking someone out accidentally if the threshold is too sensitive. Highlighting this challenge and need for human moderation, P03 voiced *‘I don’t think it should be trusted 100% with how they currently were. You should always have skepticism basically’*. Thus, although there is a strong need for moderation in social VR spaces, there needs to be a balance where people are not excluded unnecessarily.

3.4 Content Generator

Participants mentioned that AI agents could be content generators—streamers, photographers, DJs, or even business partners. Participants argued that AI agents could supplement traditional entertainment modes, e.g., gaming NPCs or streaming companions: *‘the AI on stream talk and interact with it while they’re playing a game, or even have it interact with the audience’* (P06). P01 mentioned that the AI agent could, in and of itself, be entertaining, remembering some funny conversations with the AI agent ‘Joken’. Although DJ was initially identified by P02 as a role for AI agents, they later reflected that *‘there are a lot of people who want to play music and be a DJ, so there is no need for something like that’*, highlighting the need to protect roles that people want to do in social VR from being replaced by AI agents.

3.5 Self-development and Learning

Participants thought AI agents could act as a therapist, an AI learning tool, a language learning tutor, and a social skills trainer. P03 elaborated that an AI’s ability to react non-judgmentally and with

kindness can make it more approachable, observing that socially awkward players are *‘more likely to talk to the bot than they are to anyone else around them’*. However, P06 brought up an important limitation: *‘it will not help teach somebody how to appropriately talk to somebody who doesn’t want to deal with their crap because AI is always happy to talk to you’*. Overall, participants felt AI agents could be patient teachers and help people learn skills, but AI’s positive tone and personality can lead to unrealistic responses that may not translate to human-human interactions.

4 CONCLUSION

Our work extends prior research that traces the shift in AI from rule-based to neural architectures [8] by centering specifically on LLM-driven agents. Our findings mirror some of the most recent uses of AI agents in social VR (e.g., educational applications [4, 7] and games [9]). Whereas earlier studies have emphasized the advantages of empathic agents [8], our study cautions that a model’s unrelenting positivity may foster unrealistic expectations. We also surfaced directions for future work in accessible social VR experiences and self-development. Our subsequent steps include a deeper investigation into users’ and developers’ anxieties, concerns, and requirements regarding AI agents in social VR. We further aim to identify which characteristics of AI agents in social VR are considered desirable, as well as the current obstacles to their deployment and long-term upkeep, in ways that respect users’ wishes to sustain social bonds and human creativity.

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REFERENCES

- [1] E. Bozkir, S. Özdel, K. H. C. Lau, M. Wang, H. Gao, and E. Kasneci. Embedding large language models into extended reality: Opportunities and challenges for inclusion, engagement, and privacy. In *Proceedings of the 6th ACM Conference on Conversational User Interfaces*. ACM, NY, USA, 2024. doi: 10.1145/3640794.3665563 1
- [2] V. Braun, V. Clarke, and N. Rance. How to use thematic analysis with interview data. *The counselling & psychotherapy research handbook*, 3:183–197, 2014. 1
- [3] J. McVeigh-Schultz, A. Kolesnichenko, and K. Isbister. Shaping pro-social interaction in vr: an emerging design framework. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, pp. 1–12, 2019. 1
- [4] M. Pan, A. Kitson, H. Wan, and M. Prpa. Ellma-t: an embodied llm-agent for supporting english language learning in social vr. In *Proceedings of the 2025 ACM Designing Interactive Systems Conference*, pp. 576–594, 2025. 1, 2
- [5] M. Prpa, G. Troiano, B. Yao, T. J.-J. Li, D. Wang, and H. Gu. Challenges and opportunities of llm-based synthetic personae and data in hci. In *Companion Publication of the 2024 Conference on Computer-Supported Cooperative Work and Social Computing*, p. 716–719. ACM, NY, NY, USA, 2024. doi: 10.1145/3678884.3681826 1
- [6] M. Rzeszewski, T. Osborne, P. Jones, L. Evans, and G. Weitkamp. Interviewing in the metaverse: The renewed importance of location and embodiment. *Applied Geography*, 167:103295, 2024. 1
- [7] Y. Tang, J. Situ, A. Y. Cui, M. Wu, and Y. Huang. Llm integration in extended reality: A comprehensive review of current trends, challenges, and future perspectives. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. ACM, New York, NY, USA, 2025. doi: 10.1145/3706598.3714224 1, 2
- [8] F.-C. Yang, P. Acevedo, S. Guo, M. Choi, and C. Mousas. Embodied conversational agents in extended reality: A systematic review. *IEEE Access*, 2025. 1, 2
- [9] S. Özkaya, S. Berrezueta-Guzman, and S. Wagner. How llms are shaping the future of virtual reality. *IEEE Access*, 13:193335–193355, 2025. doi: 10.1109/ACCESS.2025.3631594 1, 2